Lesson plan · for the teacher

Keep, archive, delete

Data engineering · Level 4 · Lesson 4

Time
40 min

Materials
Record cards: registers, test papers, sports-day photos, report cards, trip forms; Your ministry's or school's retention rules, if there are any; Optional: a phone or computer for the SQL step

Learning goals

- Give every record a keep-until rule.
- Back up with 3-2-1, and test a restore.
- Archive in open formats, and delete on purpose.

Activity: Keep or delete?

1. Warm-up (5 min): Ask what old records the school still holds, on paper and on computers, and who decided to keep them.
2. Teach (8 min): Explain keep-until rules, 3-2-1 backups with a restore test, open formats, and deleting on purpose, backup copies included.
3. Sort (12 min): Groups take the record cards and decide for each how long it's needed, why, and who should delete it.
4. Compare (10 min): Compare the groups' periods with the ministry's or school's real rules, where they exist. Where there are none, agree one.
5. Wrap-up (5 min): Agree who will restore one file from the backup this term, and when. Then give the exit ticket.

Find your ministry's retention rules before the lesson, if they exist. Use cards that name kinds of records, never real pupils' documents.

Discussion

- Which record should we keep the longest, and why?
- Where is our away-from-school copy?
- How do we know our backup works?

Exit ticket

- Write one record, its keep-until rule, the reason, and who deletes it.

Try it at school tomorrow
Play "Keep or delete?": give groups cards of school records (registers, test papers, sports-day photos, report cards). For each, they decide how long it's needed and why. Then compare with your school's real rules.

Practice questions · for you

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1 **The marks are copied to a USB drive every week, but no copy has ever been opened. What do you really have?**

◯ A hope: restore a file and open it to be sure ◯ A tested backup ◯ Three copies

2 **Which of these belong in a row of the school's retention table? Choose all that apply.**

☐ Why it is kept that long ☐ Who deletes it ☐ How long it is kept ☐ The kind of record

☐ The names of the pupils in it

3 **Make this term's deletion list: the id and kind of every record whose year plus its years to keep is before 2026. The rules table says how long each kind is kept.**

```
SELECT records.id, records.kind FROM records
WHERE records.year < 2026;
```

Write your program:

4 **Fill the gaps: once the list has been checked, delete the registers kept for more than 5 years.**

```
_____ FROM records
WHERE kind = 'register' AND year + 5 _____ 2026;
```

Word bank: `SELECT` `DELETE` `>` `<`

5 **Files deleted from an old school computer can often be recovered, even after the bin has been emptied.**

◯ True ◯ False

✎ **Which records does your school keep that nobody decided to keep? What keep-until rule would you give them?**

Check your answers

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1 **A hope: restore a file and open it to be sure**

Until you've restored a file and opened it, you don't know the backup works.

2 **The kind of record; How long it is kept; Why it is kept that long; Who deletes it**

A retention rule names the record, the period, the reason and who acts.

3

```
SELECT records.id, records.kind FROM records
JOIN rules ON rules.kind = records.kind
WHERE records.year + rules.keep_years < 2026;
```

The join brings each record's rule next to it, so the query can add the years: the 2018 and 2020 registers (kept 5 years), the 2024 test paper and the 2023 sports photo (kept 1 year) are due.

4

```
DELETE FROM records
WHERE kind = 'register' AND year + 5 < 2026;
```

DELETE removes the matching rows, so run the same WHERE with SELECT first and check the list.

5 **True**

That is why deleting files isn't enough.